

## EUGENE, OR, USA

WMO: 726930

Lat: 44.128N

Lon: 123.221W

Elev: 353

StdP: 14.51

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24221

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | 23.7       | 27.2 | 17.5                          | 13.5 | 27.0 | 22.6 | 17.2 | 29.4 | 24.8                  | 49.1 | 21.7 | 48.4 | 4.3                   | 0    | 0.546 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 30.2                   | 92.2            | 66.7 | 88.3 | 65.7 | 84.6 | 64.4 | 68.7                | 87.3 | 67.1 | 85.0 | 65.5 | 82.1 | 11.0                 | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.0                            | 84.1 | 74.3 | 60.2 | 79.0 | 72.4 | 58.7 | 74.7 | 70.6 | 32.9          | 87.8 | 31.6 | 84.8 | 30.4 | 82.0 | 76.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |     |            |     |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-----|------------|-----|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |     | n=20 years |     | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max | Min        | Max | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |     |
| 19.6              | 17.4 | 15.7 | DB                         | 17.8 | 99.9               | 7.0 | 3.3                                                | 12.8 | 102.3      | 8.7 | 104.2      | 4.8 | 106.1      | -0.3 | 108.5 | (4) |
|                   |      |      | WB                         | 16.8 | 72.7               | 6.9 | 1.8                                                | 11.8 | 74.0       | 7.8 | 75.0       | 3.9 | 76.0       | -1.1 | 77.3  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 53.1  | 41.4  | 43.2  | 46.8  | 50.3  | 55.8  | 60.9  | 68.1  | 68.0  | 62.8  | 53.2  | 45.6  | 40.6 | (6)  |
|      |                                                                   | DBStd       | 11.08 | 6.28  | 5.53  | 5.06  | 4.92  | 5.44  | 5.35  | 5.04  | 4.66  | 5.09  | 5.41  | 6.29  | 6.92 | (7)  |
|      |                                                                   | HDD50       | 1144  | 271   | 199   | 126   | 54    | 11    | 0     | 0     | 0     | 28    | 157   | 298   |      | (8)  |
|      |                                                                   | HDD65       | 4629  | 730   | 611   | 564   | 440   | 289   | 146   | 23    | 22    | 101   | 366   | 581   | 756  | (9)  |
|      |                                                                   | CDD50       | 2287  | 6     | 8     | 27    | 64    | 190   | 328   | 560   | 558   | 384   | 128   | 27    | 7    | (10) |
|      |                                                                   | CDD65       | 298   | 0     | 0     | 0     | 0     | 4     | 24    | 118   | 116   | 35    | 1     | 0     | 0    | (11) |
|      |                                                                   | CDH74       | 4980  | 0     | 0     | 1     | 16    | 144   | 509   | 1893  | 1691  | 695   | 31    | 0     | 0    | (12) |
|      |                                                                   | CDH80       | 1981  | 0     | 0     | 0     | 0     | 26    | 172   | 830   | 711   | 238   | 4     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 6.7   | 6.6   | 6.9   | 7.2   | 6.8   | 6.2   | 6.6   | 7.2   | 6.8   | 6.5   | 6.2   | 6.6   | 6.9  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 47.8  | 7.4   | 5.3   | 5.5   | 3.3   | 2.4   | 1.3   | 0.5   | 0.9   | 1.4   | 3.1   | 8.0   | 8.1  | (15) |
| (16) |                                                                   | PrecMax     | 64.0  | 15.1  | 14.2  | 12.4  | 7.8   | 6.9   | 3.9   | 3.0   | 5.8   | 4.6   | 8.1   | 20.5  | 21.0 | (16) |
| (17) |                                                                   | PrecMin     | 27.7  | 0.3   | 0.9   | 0.8   | 0.5   | 0.2   | 0.1   | 0.0   | 0.0   | 0.1   | 1.3   | 1.3   |      | (17) |
| (18) |                                                                   | PrecStd     | 9.1   | 3.9   | 3.1   | 2.8   | 1.9   | 1.5   | 1.0   | 0.6   | 1.2   | 1.1   | 2.0   | 4.1   | 4.8  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 59.4  | 61.7  | 69.1  | 76.4  | 83.3  | 92.0  | 98.9  | 96.3  | 91.5  | 77.6  | 65.8  | 59.1  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 54.7  | 54.4  | 56.3  | 62.5  | 66.4  | 68.7  | 68.8  | 67.1  | 64.3  | 60.2  | 59.1  | 53.9  |      | (20) |
| (21) |                                                                   | 2% DB       | 55.7  | 57.3  | 63.5  | 70.1  | 78.1  | 84.7  | 92.8  | 91.7  | 86.4  | 72.5  | 60.5  | 55.6  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 52.1  | 51.4  | 54.0  | 58.6  | 63.9  | 65.1  | 67.3  | 66.1  | 63.5  | 57.9  | 55.6  | 52.4  |      | (22) |
| (23) |                                                                   | 5% DB       | 53.0  | 54.4  | 59.5  | 65.2  | 73.4  | 79.7  | 88.5  | 87.7  | 82.1  | 67.6  | 57.4  | 52.7  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 50.1  | 49.9  | 52.3  | 55.4  | 61.5  | 63.4  | 65.9  | 65.5  | 62.6  | 56.7  | 53.8  | 50.3  |      | (24) |
| (25) |                                                                   | 10% DB      | 50.7  | 51.9  | 56.1  | 60.7  | 68.6  | 75.1  | 84.4  | 83.6  | 77.3  | 63.6  | 54.8  | 50.1  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 48.1  | 48.2  | 50.4  | 52.8  | 58.5  | 61.1  | 64.6  | 64.2  | 60.9  | 55.9  | 51.7  | 47.7  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 56.2  | 56.4  | 57.9  | 63.8  | 68.4  | 69.9  | 72.1  | 69.8  | 68.1  | 63.8  | 60.2  | 55.5  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 58.3  | 60.2  | 65.6  | 74.6  | 80.6  | 87.9  | 93.3  | 90.3  | 85.1  | 72.4  | 64.0  | 57.5  |      | (28) |
| (29) |                                                                   | 2% WB       | 52.8  | 53.0  | 55.2  | 59.3  | 64.6  | 66.7  | 69.0  | 68.0  | 65.6  | 60.6  | 57.0  | 52.8  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 55.0  | 55.6  | 61.7  | 68.6  | 76.0  | 82.9  | 88.3  | 87.8  | 81.1  | 67.9  | 59.6  | 54.8  |      | (30) |
| (31) |                                                                   | 5% WB       | 50.5  | 50.6  | 53.0  | 56.1  | 61.9  | 64.2  | 67.1  | 66.4  | 63.9  | 58.6  | 54.3  | 50.6  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 52.7  | 53.6  | 58.1  | 63.7  | 72.4  | 78.1  | 85.8  | 85.0  | 78.8  | 64.6  | 56.7  | 52.7  |      | (32) |
| (33) |                                                                   | 10% WB      | 48.3  | 48.4  | 51.1  | 53.4  | 59.2  | 61.9  | 65.3  | 64.7  | 62.0  | 56.7  | 52.1  | 47.9  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 50.5  | 51.4  | 55.4  | 59.3  | 67.9  | 73.9  | 82.7  | 81.4  | 74.9  | 62.1  | 54.3  | 50.0  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.5  | 14.4  | 17.1  | 19.2  | 21.8  | 24.5  | 30.2  | 30.0  | 27.3  | 20.4  | 13.4  | 10.8  |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 13.9  | 17.8  | 23.4  | 28.4  | 31.6  | 34.0  | 37.5  | 37.5  | 36.8  | 29.0  | 15.7  | 12.3  |      | (36) |
| (37) |                                                                   | 5% DB MCWBR | 10.3  | 12.3  | 14.8  | 17.9  | 18.7  | 16.5  | 15.1  | 15.2  | 16.5  | 16.1  | 10.9  | 9.6   |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 12.9  | 15.3  | 21.6  | 26.2  | 29.9  | 31.8  | 35.0  | 34.2  | 31.9  | 21.6  | 13.2  | 11.3  |      | (38) |
| (39) |                                                                   | 5% WB MCWBR | 10.2  | 11.0  | 14.2  | 17.0  | 18.1  | 16.1  | 15.3  | 14.7  | 15.0  | 13.1  | 10.0  | 9.3   |      | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.313 | 0.315 | 0.318 | 0.355 | 0.358 | 0.355 | 0.349 | 0.355 | 0.344 | 0.323 | 0.317 | 0.322 |      | (40) |
| (41) |                                                                   | taud        | 2.554 | 2.558 | 2.558 | 2.418 | 2.430 | 2.442 | 2.478 | 2.481 | 2.497 | 2.569 | 2.553 | 2.500 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 260   | 281   | 294   | 288   | 289   | 290   | 290   | 285   | 280   | 274   | 256   | 243   |      | (42) |
| (43) |                                                                   | Edh at Noon | 21    | 25    | 28    | 35    | 36    | 35    | 34    | 32    | 30    | 24    | 21    | 20    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 436   | 690   | 1007  | 1437  | 1826  | 2038  | 2307  | 1986  | 1494  | 892   | 488   | 351   |      | (44) |
| (45) |                                                                   | RadStd      | 64    | 90    | 121   | 121   | 171   | 173   | 133   | 117   | 121   | 95    | 61    | 53    |      | (45) |

## Historical Trends

|              |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |
|--------------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|
|              |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |
|              |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |
| Station Only |                         | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | +43 |
|              | Regional (25 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +166  | +54 |

Nomenclature: See separate page